

# Julián David Cabrera Barragán

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## PROJECTS

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### Microenterprise Density Prediction System (GoDaddy)

2025

*Python - Machine Learning - Time Series - Simulation*

Academic

- Analysis and prediction of microbusiness density using the GoDaddy dataset (Kaggle).
- Implementation of Machine Learning models in Python (Random Forest).
- Data preprocessing, temporal feature engineering, and model validation.
- Model evaluation using RMSE, MAE, and drift detection techniques.

### Hotel Management Database System

2025

*PostgreSQL - SQL - Relational Model - Triggers*

Academic

- Design of a relational database for a hotel management system.
- Definition of tables, primary and foreign keys, and data normalization.
- Implementation of business rules using triggers and PL/pgSQL functions.
- Automatic validation of room availability and service usage.

### Music Management System

2025

*C++ - AVL Trees - Data Structures*

Academic

- Development of a CRUD system for albums, songs, and artists.
- Search optimization using AVL Trees combined with lists.
- Data persistence using text files.

### CPU Scheduling Simulator

2023

*JavaScript - FCFS - SJF - SRTF - Round Robin*

Academic

- Simulation of CPU scheduling algorithms.
- Computation of performance metrics: turnaround, waiting, and response time.

### Cultivable Land Prediction System

2024

*Python - Decision Trees - AI*

Academic

- Classification model to predict soil suitability for agriculture.
- Use of environmental and agricultural variables for decision support.

## EDUCATION

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### Francisco José de Caldas District University

In Progress

*B.Sc. in Systems Engineering*

Bogotá, Colombia

### SENA

2025

*Course: Software Quality Application in the Development Process*

## TECHNICAL SKILLS

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**Programming Languages:** Python, C++, JavaScript, Java

**Databases:** PostgreSQL, MongoDB, MariaDB

**SQL:** Advanced queries, normalization, foreign keys, triggers, functions

**Machine Learning:** Random Forest, Decision Trees, data preprocessing

**Tools:** Git, GitHub, HTML, CSS